F5 · First 3 held-out prompts (fixed rule), steered with the D2 direction (content) at c = ±4 · Qwen2.5-3B-Instruct · greedy, 128 tokens, text truncated for display

Prompt: Describe the planets of the solar system.

c = -4

Certainly! The solar system consists of the Sun at its center and eight major planets that orbit around it. These planets are classified into two main groups: the inner terrestrial planets and the outer gas giants. Here's a detailed description of each planet: ### 1. Mercury - **Size**: Mercury is the smallest planet in our solar system. - **Distance from the Sun**: It is the closest planet to the Sun, orbiting at a […]

c = +0 (unsteered)

Certainly! The solar system consists of the Sun at its center and eight major planets that orbit around it. Here is a detailed description of each planet in our solar system: 1. **Mercury**:   - **Distance from the Sun**: 57.9 million km (35.9 million miles)   - **Orbital Period**: 88 Earth days   - **Diameter**: 4,879 km (3,032 miles)   - **Mass**: 3.3 x 10²³ kg   - **Density**: 5.427 g

c = +4

Certainly! I'll provide a brief description of each planet in our solar system, starting from the one closest to the Sun and moving outward. 1. **Mercury**: Mercury is the closest planet to the Sun and the smallest in size among the terrestrial planets. It has no moons and lacks an atmosphere. Due to its proximity to the Sun, temperatures on its surface can range from extremely hot during the day (around 430°C) to v […]

Prompt: Explain what a black hole is.

A black hole is a region of spacetime where the gravitational force is so strong that nothing, not even light, can escape from it once it crosses a certain boundary known as the event horizon. This boundary is defined by the Schwarzschild radius, which depends on the mass of the black hole. ### Key Components of a Black Hole: 1. **Event Horizon**: This is the point of no return around a black hole. Any object or ra […]

A black hole is a region of space where the gravitational pull is so strong that nothing, not even light, can escape from it once it crosses a certain boundary known as the event horizon. This makes black holes fundamentally invisible to direct observation. ### Key Characteristics of Black Holes: 1. **Gravitational Field**: The gravitational field around a black hole is extremely strong. It warps spacetime in such […]

A black hole is one of the most fascinating and mysterious objects in the universe. It is a region of space where the gravitational pull is so strong that nothing, not even light, can escape from it once it crosses a certain boundary known as the event horizon. Here's an explanation of what a black hole is: 1. **Formation**: Black holes typically form when massive stars collapse under their own gravity after they ha […]

Prompt: Describe how the Moon affects ocean tides.

The Moon's gravitational pull is the primary driver of ocean tides on Earth. The gravitational force exerted by the Moon causes the water in the oceans to bulge out in two places: one directly towards the Moon and one directly opposite it. This phenomenon is known as the tidal bulge. Here’s a detailed breakdown of how this works: 1. **Gravitational Attraction**: The Moon has a significant gravitational field that a […]

The Moon's gravitational pull is the primary cause of ocean tides on Earth. Here’s a detailed explanation of how this works: 1. **Gravitational Attraction**: The Moon exerts a gravitational force on the Earth. This force is strongest at the point closest to the Moon (the near side) and weakest at the opposite point (the far side). However, because the Earth is not a perfect sphere but an oblate spheroid (it bulges s […]

The Moon's gravitational pull is the primary cause of ocean tides on Earth. Here’s a detailed explanation of how this works: 1. **Gravitational Attraction**: The Moon exerts a gravitational force on the Earth, and this force is strongest on the side of the Earth closest to the Moon. This gravitational attraction pulls on the water in the oceans, causing it to bulge out towards the Moon. 2. **Earth-Moon Distance**: […]